INCH-POUND

MIL-C-24643/50C
22 November 1994
SUPERSEDING
MIL-C-24643/50B
26 March 1991

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, FOUR CONDUCTOR, TYPE LSFNW (INCLUDING VARIATION LSFNWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LSFNW (unarmored) and LSFNWA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated (see table I for size).
Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Crosslinked polyethylene insulation or ethylene propylene rubber insulation (see table I for thickness) Standard identification code shall be applied by method 1 or method 3. If identification method 1 is used, insulation color
* shall be black.
Fourth - Four conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly.
Fifth - Binder tape applied helically with overlap.

* Construction variant for type LSFNW

* Sixth - Cross-linked polyolefin jacket. (see table I for thickness.)

Construction variant for type LSFNWA

* Sixth - Cross-linked polyolefin jacket. (see table I for thickness.)
Seventh - Braided metal armor.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Details.

Military 1/ part no. M24643/50	Size 2/	AWG conductor size	Insulation thickness min avg (inch)	Cable jacket thickness min avg (inch)	Overall diameter (inches)			
					TYPE LSFNW		TYPE LSFNWA	
					min	max	min	max
-01UN	3	16 (class B)	0.030	0.040	0.413	0.447	0.463	0.497
-02UN	4	14 (class B)	.030	.040	.475	.513	.525	.563
-03UN	9	10 (class B)	.030	.040	.583	.630	.635	.680
-04UN	23	7 (class B)	.045	.050	.768	.830	.818	.880

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSFNW and LSFNWA.

TABLE II. Electrical details.

Size 1/	AWG conductor size	Conductor resistance (ohms)	Insulation resistance (megohms)	Voltage withstand (minimum) (volts)	Cold Bending mandrel diameter (max inch)	Accelerated service loading current (amperes)
3	16 (class B)	4.3	6000	4000	4	----
4	14 (class B)	2.68	5000	4000	4	----
9	10 (class B)	1.08	3800	4000	5	----
23	7 (class B)	.543	3700	5500	8	115

1/ Type LSFNW and LSFNWA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, max	(see table II)
Voltage withstand - volts, root mean square, minimum	(see table II)
Insulation resistance - Conductor to conductor Megohms/1000 feet minimum	(see table II)
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Cold bending, cable - (see table I for mandrel diameter)	No damage
Tear - pounds per inch thickness, minimum (ASTM D 470)	35

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	Requirements
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene	
Air Oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged minimum	75
Cross-linked polyethylene	
Air Oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (conductor) - method 1 only, cycles, minimum	25
*Permanence of printing (jacket) - cycles, minimum	125
Armor (type LSFNWA only) - conformance to material, construction and coverage	No failure
Group D:	
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:	Requirements
Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:	
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and Size	Feet (nominal)
LSFNW-3	2500
LSFNW-4, 9	2000
LSFNW-23	1500

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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